INCH-POUND

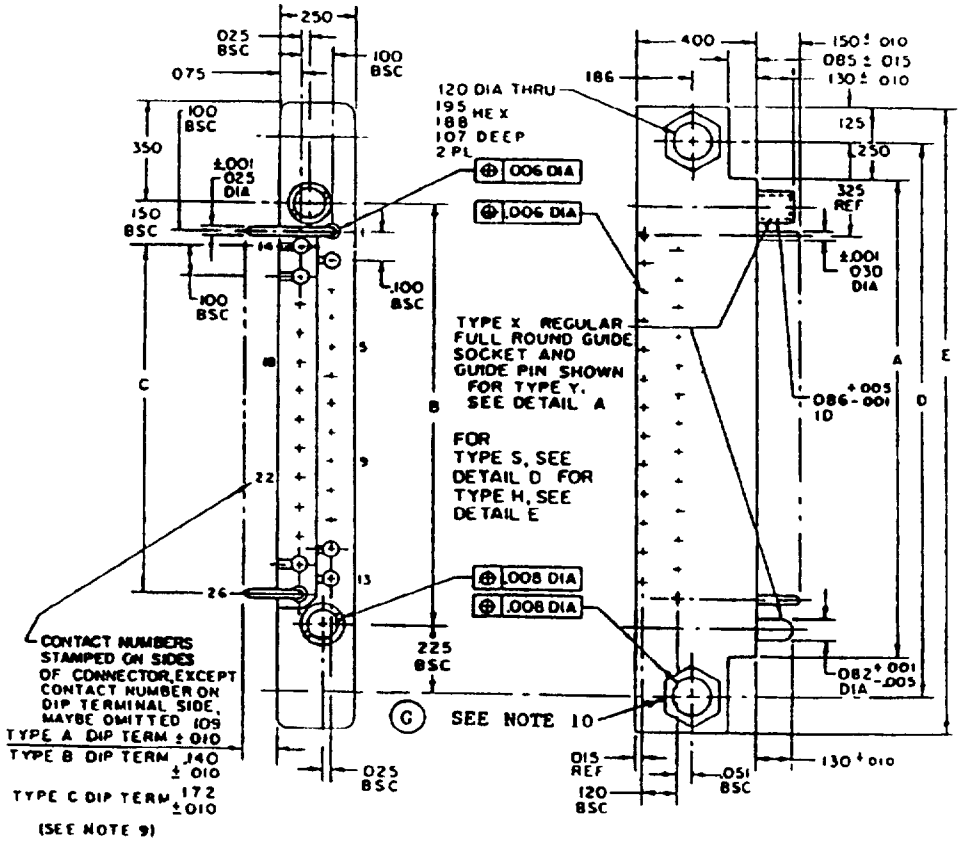
MIL-C-55302/57G
23 September 1992
SUPERSEDING
MIL-C-55302/57F
26 August 1988

MILITARY SPECIFICATION SHEET

CONNECTORS, PRINTED CIRCUIT SUBASSEMBLY AND ACCESSORIES, PLUG, PIN CONTACTS, DECADE INCREMENTS 10 THRU 70 CONTACT POSITIONS AND INTERMEDIATE POSITIONS OF 14, 24, 44, 54 AND 26, 36, 56, 66 FOR PRINTED WIRING BOARDS (100 INCH SPACING)

This specification is approved for use by all Departments and Agencies of the Department of Defense

The requirements for aquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-55302

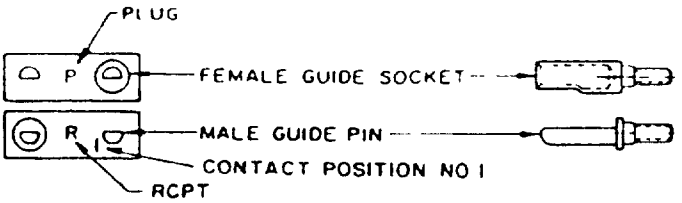


G FIGURE 1 Connector, plug .100 (2.54 mm) spacing

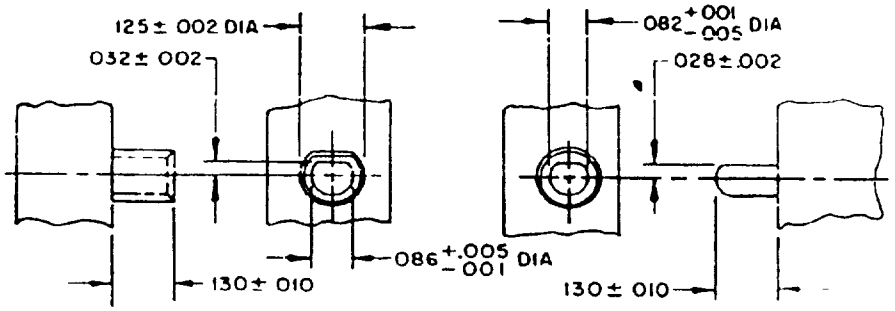
G denotes changes

DISTRIBUTION STATEMENT A Approved for public release, distribution is unlimited

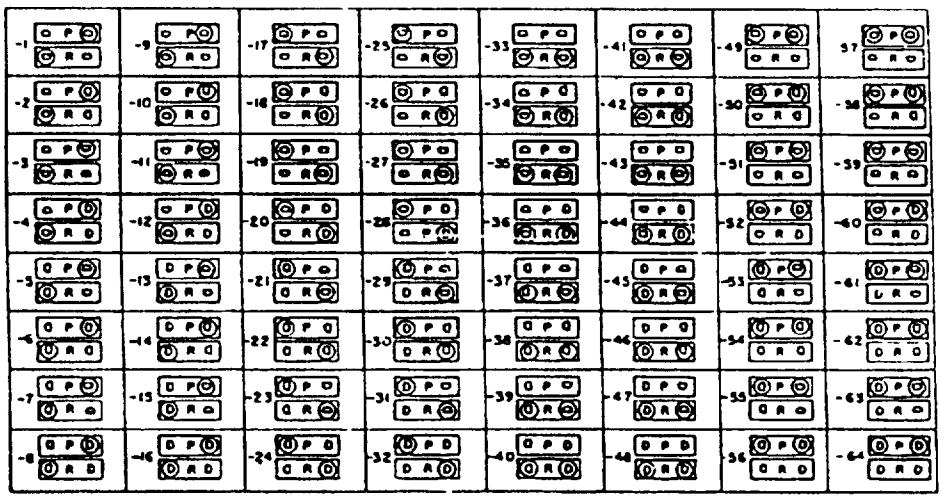
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CONNECTOR POLARIZATION WITH D SHAPED GUIDE PINS AND SOCKETS (TYPE Y)



Detail A Polarization pins/socket (For polarization configuration see detail B)

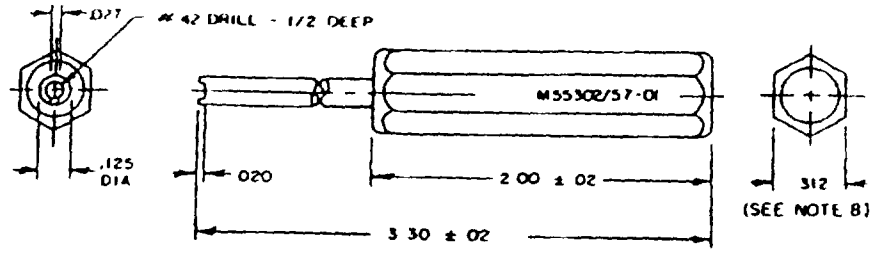


Detail B Polarization configuration chart

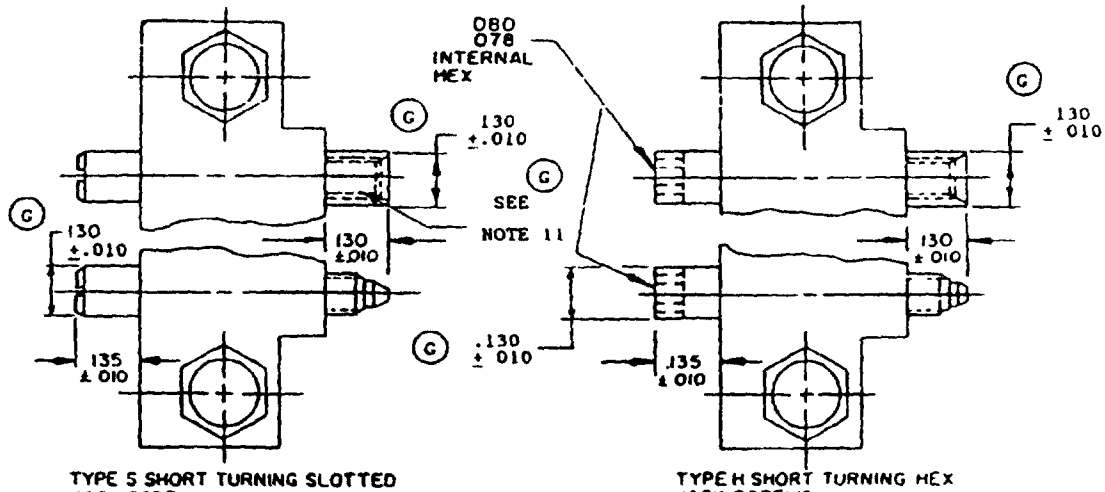
NOTE Pins/socket in chart can be easily adjusted to any desired configuration by the spanner wrench (see detail C)

FIGURE 1 Connector, plug 100 (2 54 mm) spacing - Continued

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Detail C spanner wrench



Detail D

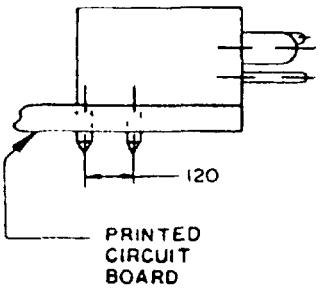
Detail E

NOTES

1. Dimensions are in inches
2. Metric equivalents are given for general information only
3. Unless otherwise specified, tolerance is ±.005 (0.13 mm)
4. These connectors mate with connectors specified in MIL-C-55302/58, MIL-C-55302/62, MIL-C-55302/64, MIL-C-55302/65, and MIL-C-55302/66 using appropriate hardware
5. Numbers indicating the first and last position in each row and every fourth contact position in between shall be marked on the sides of the connector. As an option to the above, numbers indicating every fourth cavity may be stamped on the side of the connector, with the exception that the number one contact shall be marked.
6. Termination layout on .020 (0.51 mm) modular grid
7. Metric equivalents are in parentheses.
8. Handle shape optional
9. Optional undercut .050 (1.27 mm) maximum, on PCB side for cleaning purposes
10. (G) Position of the hex orientation is optional. Hex shall not break through the side of the insulator
11. (G) Hole depth to female guide hardware is .282 min (7.16 mm). Full thread depth to female threaded hardware is .240 min (6.1 mm)

FIGURE 1 Connector, plug .100 (2.54 mm) spacing - Continued

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Inches	mm	Inches	mm	Inches	mm
001	0 03	020	0 51	075	1 91
002	0 05	025	0 64	078	1 98
005	0 13	027	0 69	.082	2 08
006	0 15	028	0 71	086	2 18
008	0 20	030	0 76	.094	2.39
.010	0 25	032	0 81	.100	2.54
015	0 38	051	1 30	107	2 72

Number of contacts	Dimensions				
	A ± 010	BSC B	REF C	BSC D	E ± 010
10	.850 (21 59)	.650 (16 51)	400 (10 16)	1.100 (27.94)	1 350 (34 29)
14	1.050 (26.67)	850 (21 59)	.600 (15 24)	1.300 (33 02)	1 550 (39 37)
20	1.350 (34.29)	1.150 (29.21)	.900 (22 86)	1.600 (40.64)	1.850 (46.99)
24	1.550 (39 37)	1.350 (34.29)	1.100 (27.94)	1.800 (45 72)	2.050 (52.97)
26	1.650 (41 91)	1.450 (36.83)	1.200 (30.48)	1.500 (48.26)	2 150 (54.61)
30	1.850 (46.99)	1.650 (41.91)	1.400 (35.56)	2.100 (53.34)	2.350 (59.69)
36	2.150 (54.61)	1.950 (49.53)	1.700 (43.18)	2 400 (60 96)	2.650 (67 31)
40	2 350 (56.69)	2 150 (54.61)	1.900 (48.26)	2 600 (66 04)	2 850 (72 39)
44	2.550 (64 77)	2.350 (56 69)	2 100 (53.35)	2 800 (71 12)	3 050 (77 47)
50	2.850 (72.39)	2.650 (67 31)	2.400 (60 96)	3.100 (78 74)	3.350 (85.09)
54	3.050 (77 47)	2.850 (72.39)	2.600 (66 04)	3.300 (83.82)	3.550 (90.17)
56	3 150 (80.01)	2.950 (74 93)	2.700 (68.58)	3.400 (86.36)	3.650 (92 71)
60	3 350 (85 09)	3 150 (80 01)	2.900 (73.66)	3.600 (91 44)	3.850 (97 79)
66	3 650 (92 71)	3.450 (87.63)	3.200 (81.28)	3 900 (99 06)	4 150 (105 41)
70	3 850 (97 79)	3.650 (92.71)	3.400 (86.36)	4 100 (104 14)	4.350 (110 49)

FIGURE 1 Connector, plug 100 (2 54) spacing - Continued

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REQUIREMENTS

Dimensions and configuration See figure 1

Material Guide pins, type X guide sockets, jackscrews and jacksockets shall be made of corrosion resisting stainless steel in accordance with ASTM-A581 or ASTM-A582, passivated in accordance with QQ-P-35 or shall be corrosion resistant copper nickel alloy (61 ±2% copper, 25 ±2% nickel, 2 5% maximum other, and the balance zinc) and shall have a brinell hardness of 145 to 175 Type Y guide sockets shall be beryllium copper in accordance with QQ-C-550, nickel plated in accordance with QQ-N-290, class 2, 50 to 150 microinches

Plating. The contact plating shall be in accordance with MIL-C-55302 or gold in accordance with MIL-G-45204, type II, grade C, class 1, over nickel plating in accordance with QQ-N-290, class 2, 50 to 150 microinches in the engaging area and gold in accordance with MIL-G-45204, type II, grade C, class 00, or type III, grade A, class 00, over nickel plating in accordance with QQ-N-290, class 02, 50 to 150 microinches in the termination area

Contact identification· See figure 1

Contact resistance The average contact resistance of all contacts measured shall not exceed 0.010 ohm, and no individual contact pair shall have a resistance exceeding 0 020 ohm

Dielectric withstanding voltage

Sea level 1,000 volts rms.
High altitude 300 volts rms

Current rating 5.0 amperes maximum in accordance with contact.

Polarization Type X, regular, guide pin and socket, full round (see figure 1) Type Y, adjustable "D" shaped guide pin and/or socket (see figure 1, details A and B). All connectors are supplied in the -1 polarized position For other than OEM use, connectors may be repolarized by using M55302/57-01 spanner wrench and M55302/57-02 guide pin and guide socket kit. When guides are removed and reinstalled in another position apply Loctite 83-CV or equivalent to the threads Ordering connectors in other than the -1 polarized position may be done by the OEM's by placing the desired polarized position in brackets three spaces after the part number on the purchase order Example M55302/57-A10Y -32 . Polarization pins and sockets shall be adjusted to any desired configuration by spanner wrench (M55302/57-01) Ordering connectors in the -1 polarized position without Loctite 83-CV or equivalent applied may be done by placing an "-0" in brackets three spaces after the part number on the purchase order Guide sets, as installed, shall withstand 7 ounce-inches backoff torque applied through the guide set mounting nut

M55302/57-02 kit Consists of one type Y "D" shaped guide socket and one type Y "D" shaped guide pin

Example of part number

M55302/57	A	10	X
Basic number of specification sheet	Type of terminal (see figure 1)	Number of contacts (see figure 1)	Type of mounting hardware (see details A, B, C, D, E of figure 1)

A = dip terminal .109 long
B = dip terminal 140 long
C = dip terminal 172 long

Type X = full round
Type Y = "D" shaped
Type S = 135 slotted turning jackscrew
Type H = 135 hex turning jackscrew

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CONCLUDING MATERIAL

Custodians.
Army - CR
Navy - EC
Air Force - 17
(G) NASA - NA

Review activities
Army - AR, MI
Navy - AS, TD
Air Force - 85, 99
DLA - ES

User activities
Army - AT, AV, ME
Navy - MC, OS
Air Force - 11, 19, 80

Preparing activity
Army - CR

Agent.
DLA - ES

(Project 5935-3860-3)

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